AI-Based Multi-Constraint Delivery Route Optimizer

Priority-Aware, Disruption-Resilient Routing using Ant Colony Optimization

Zeroth Review Presentation

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Problem Statement

- To develop a system that efficiently plans delivery routes across multiple stops while considering real-world constraints such as vehicle capacity, delivery priorities, time windows, and dynamic road conditions.

Why is it Important?

- Inefficient routing directly increases fuel cost, delivery time, and operational expense.
- Delayed critical deliveries (medicines, blood units) can have serious real-world consequences.
- Businesses need routing systems that adapt to disruptions instead of failing silently.
- Sustainability: shorter, smarter routes reduce fuel consumption and emissions.

Who is Affected?

- **Logistics and delivery companies** – inefficient routing increases operating cost.
- **Hospitals, pharmacies, and blood banks** – rely on timely, priority-aware delivery.
- **Disaster relief organizations** – need routing that adapts to blocked/closed roads.
- **End customers** – affected by delayed or unreliable delivery windows.
- **Delivery drivers** – affected by route length, load balance, and re-planning overhead.

Proposed Solution

- A web application where delivery stops are entered with package weight, time window, and priority level.
- Routes are optimized using **Ant Colony Optimization (ACO)**, a nature-inspired algorithm, extended to respect:
  - Vehicle capacity and delivery time windows (CVRPTW)
  - Delivery priority (Critical / High / Normal / Low)
- The system **dynamically re-optimizes** routes when a road closure is reported.
- A **Genetic Algorithm** solves the same problem independently, for comparison.
- A **dashboard** summarizes optimization benefit across completed deliveries.

Objectives of the Project

1. Design and implement an ACO-based route optimizer that solves a Capacitated Vehicle Routing Problem with Time Windows (CVRPTW).
2. Incorporate delivery priority levels into the optimization cost function.
3. Enable dynamic re-optimization of routes in response to live road closures.
4. Compare ACO performance against a Genetic Algorithm on the same problem instances.
5. Provide an analytics dashboard that quantifies the benefit of optimization (distance saved, success rate).

Scope of the Project

In scope:

- Single-vehicle, single-depot routing with a configurable number of delivery stops
- Synthetic/user-entered data for stops, weights, time windows, and priorities
- Simulated road closures (manually marked, not live traffic feed)
- Comparative study between ACO and Genetic Algorithm

Out of scope:

- Real-time GPS/traffic API integration
- Multi-vehicle or multi-depot fleet assignment
- Production-grade logistics platform integration

Existing System and Its Drawbacks

Current approaches:

- Manual route planning by dispatchers based on experience
- Generic shortest-path tools (e.g., basic map routing) used for sequencing stops

Drawbacks:

- Do not account for vehicle capacity or delivery time windows
- Treat all deliveries as equally urgent – no priority handling
- Static: cannot adapt automatically when a road becomes unavailable
- No visibility into whether a route is actually close to optimal

Proposed System and Workflow

1. User enters delivery stops: location, weight, time window, priority level, and vehicle capacity.
2. System builds a distance/time matrix between all stops.
3. ACO algorithm searches for a valid, efficient route respecting capacity, time windows, and priority.
4. *(Optional)* Admin marks a road as closed → affected edge removed from the matrix → route re-optimized live.
5. Genetic Algorithm independently solves the same problem for comparison.
6. Results displayed: optimized route on map, convergence graph, algorithm comparison, analytics dashboard.

Methodology – Ant Colony Optimization

- Initialize pheromone trails uniformly across all edges (routes between stops).
- Each simulated ant constructs a candidate route by probabilistically choosing the next stop, based on:
  - Pheromone strength on the edge
  - Heuristic desirability (inverse of distance, weighted by delivery priority)
- Routes violating capacity or time-window constraints are penalized in the cost function.
- Pheromone trails are updated: reinforced on shorter/valid routes, evaporated over time on weaker ones.
- Process repeats over many iterations until the route quality converges.

Methodology – Genetic Algorithm (Comparison)

- Candidate routes encoded as chromosomes (ordered list of stops).
- Population evolves over generations using:
  - **Selection** – favor lower-cost (shorter, valid) routes
  - **Crossover** – combine segments of two parent routes
  - **Mutation** – small random reordering to maintain diversity
- Same constraint-aware cost function used as in ACO, for a fair comparison.

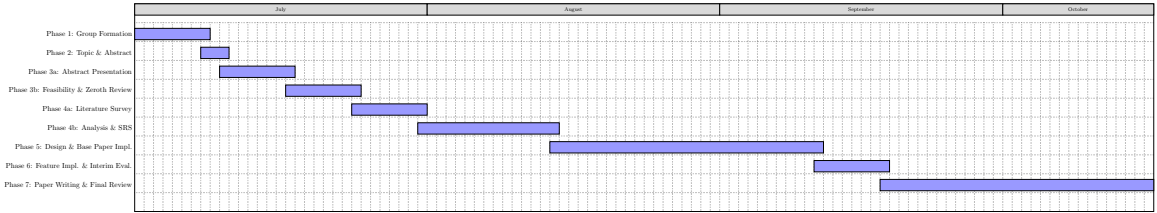
Methodology – Development Approach

- **Backend:** Python, Flask
- **Database:** SQLite/MYSQL
- **Core algorithms:** ACO and Genetic Algorithm, implemented from scratch
- **Frontend:** React.js, Leaflet.js (map), Chart.js (charts/dashboard)
- **Development process:** Iterative development aligned with project phase schedule, version control using Git

Expected Outcome

- A working web application that generates optimized delivery routes under capacity, time-window, and priority constraints.
- Demonstrated live re-routing when a road closure is introduced.
- Quantified comparison between ACO and Genetic Algorithm (route quality, convergence speed).
- Analytics dashboard showing measurable improvement over an unoptimized baseline route (distance saved, success rate).
- A base paper documenting the methodology and results, suitable for publication.

Timeline – Gantt Chart



Milestones and Status

Milestone	Target Date	Status
Group formation & team lead	08 Jul 2026	Completed
Topic identification & abstract	10 Jul 2026	Completed
Abstract presentation & guide selection	17 Jul 2026	Completed
Feasibility study & zeroth review	24 Jul 2026	**In progress**
Literature survey (12 papers)	31 Jul 2026	Upcoming
SRS report	14 Aug 2026	Upcoming
Design report & base paper implementation	11 Sep 2026	Upcoming
First interim evaluation	18 Sep 2026	Upcoming
Final report & presentation	16 Oct 2026	Upcoming

Conclusion

- Existing delivery routing methods ignore capacity, time-window, priority, and disruption constraints that are essential in real-world logistics.
- This project designs and implements a priority-aware, disruption-resilient route optimizer using Ant Colony Optimization, benchmarked against a Genetic Algorithm.
- The result is a technically substantial, demonstrable system with clear, measurable outcomes.

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